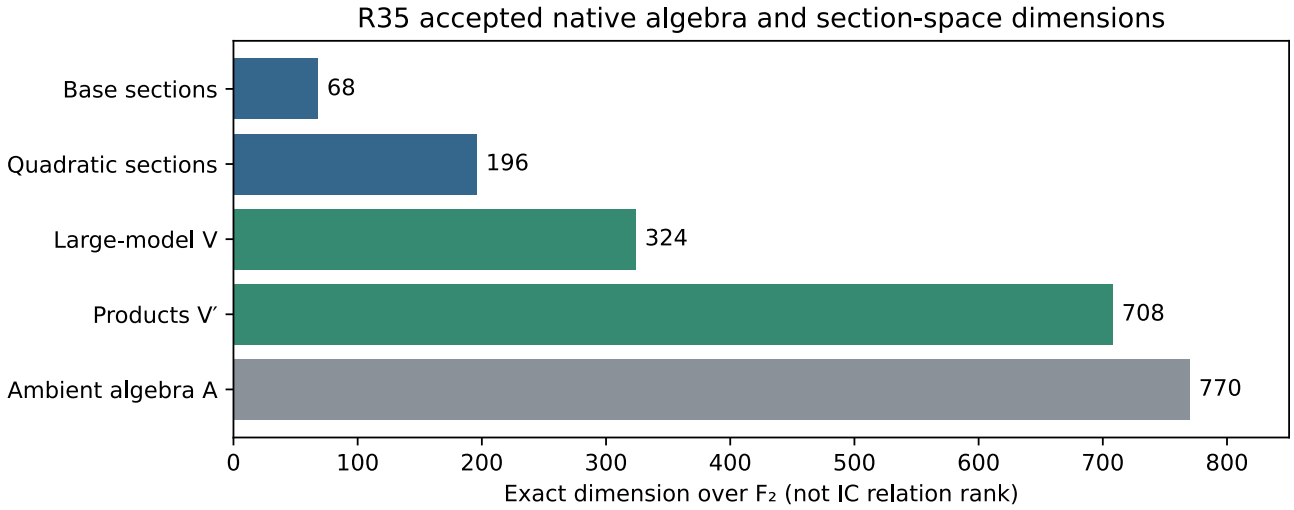
Koblitz cover transfers — R35

Exact construction observations; no IC performance claim

The fixed degree-five padding point now supplies 154 complete coefficients, a 770-dimensional algebra and exact section spaces of dimensions 68,196,324. All 770 inverse-basis columns are recombined, with the coefficient embedding tied to the actual source point.



Run algebra35-01 verifies the coefficient lift and full forward/inverse basis. Run jets35-01 checks all five source equations at every Newton stage, eight units and the discriminant, all five coefficient conjugates, 68 literal function-descent comparisons and every retained nine-term regression.

Run multiply35-01 checks all 52,650 unordered products of the 324 sections, including all products after rank 708 is reached. Scalar field convolution and binary bit-plane convolution agree on every product, and all 1,024 field products are independently checked. Every product and pivot is recombined.

Run inflate35-01 constructs all ten 196-dimensional vanishing spaces in the actual 324-dimensional ambient space, checking all 46,240 products in both arithmetic implementations and all original/ambient/new-basis recombinations. Completeness as L(3D0-D) uses the documented proper/local/pole bridge; exact native rank is not itself a proof of global geometry or a target factor map.

Lean: 4 declarations in 1 accepted pinned runs. Coefficient-lift identities use explicit characteristic-two, residue-field and nilpotence assumptions. Native arrays, normalization and Riemann–Roch are not imported as formal proofs. Every attempt is indexed in results.json.

Next: exact subspace division, flips, addition and equality, then the signed class using certified deck order and the actual F2 factor / nonzero order-11 action. Full quotient saturation, polarized embeddings, theta, node187 and larger arbitrary collector supports remain open. No timing comparison, IC relation rank, scalar recovery or attack claim. Native 4GiB/1200s limits; Lean4.34.1/mathlibd13f23b, -j1 -M4096/1200s. No independent external replay.

Reference: Khuri-Makdisi math/0409209v3, Representation B and large model; R34 LOCAL_GEOMETRY, COMPLETE_DIVISORS and NEXT_CONSTRUCTION.